

Grammar of the Wave: Towards Explainable Multivariate Time Series Event Detection via Neuro-Symbolic VLM Agents

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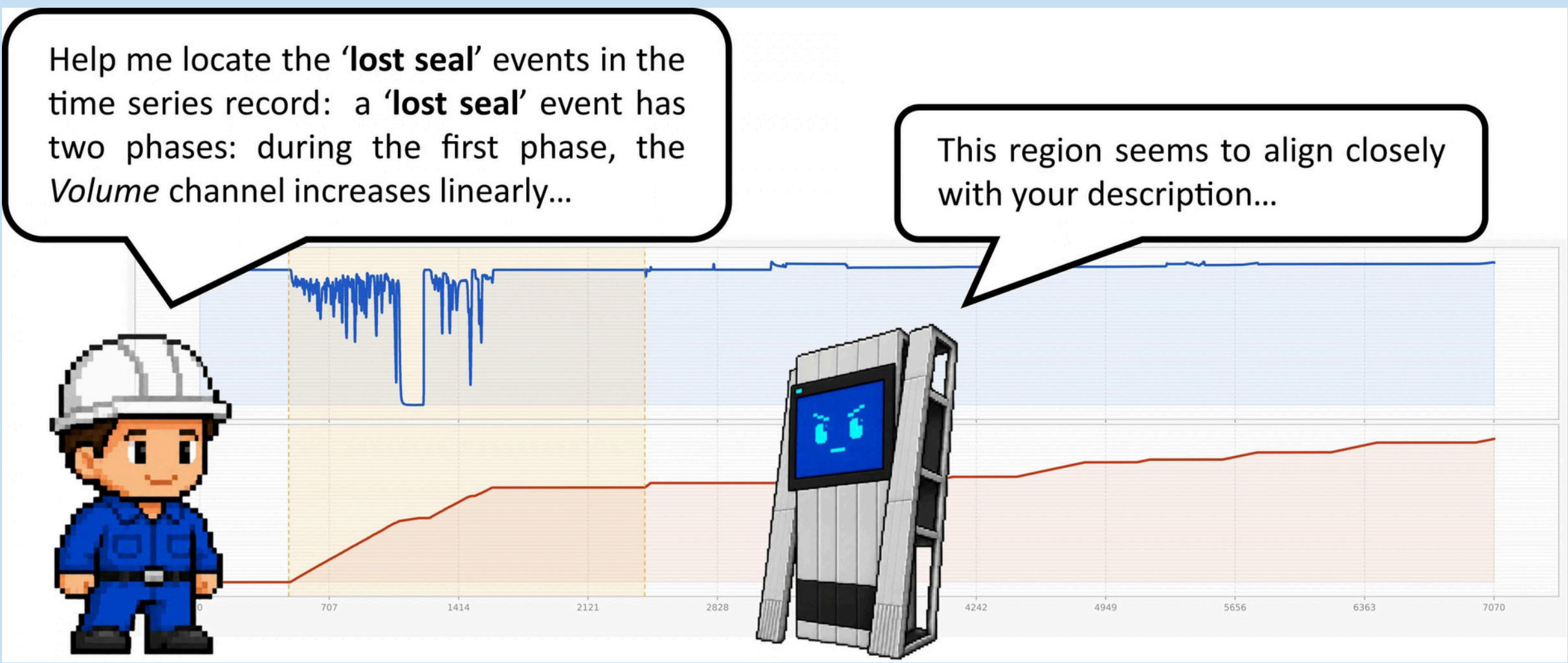
Paper

Code & Data



Can language descriptions tell us **what happens—and when**—in multivariate signals?

Yes—with explicit event logic.



ELT structure + VLM agents.

Explainable localization without training labels.

Performance

		Pressure Test macro-F1@0.5 ↑	Climate Hazard macro-F0.5 coverage ↑
CNN	Sup.	38.6 ± 2.3	40.7 ± 2.2
Trans.	Sup.	27.7 ± 2.1	40.8 ± 6.6
Timer	Sup.	32.0 ± 3.4	17.1 ± 0.6
Moment	Sup.	43.6 ± 5.4	20.4 ± 5.6
Chronos	Sup.	43.4 ± 5.6	39.9 ± 8.3
TimesFM	Sup.	46.6 ± 6.1	47.4 ± 3.4
Num.	ZS	53.2 ± 3.1	45.7 ± 3.6
Num.	FS	57.8 ± 6.0	49.4 ± 1.1
VL-T.	ZS	51.9 ± 2.1	26.0 ± 3.3
VL-T.	FS	53.6 ± 3.2	33.2 ± 4.0
SELA	ZS	66.8 ± 3.9	57.1 ± 2.2

KITE: PT 62 samples / 2 channels · CH 86 samples / 7 channels

Sup.: supervised · ZS/FS: zero/few-shot · LLM/VLM: GPT-5 · Table 4 (%)

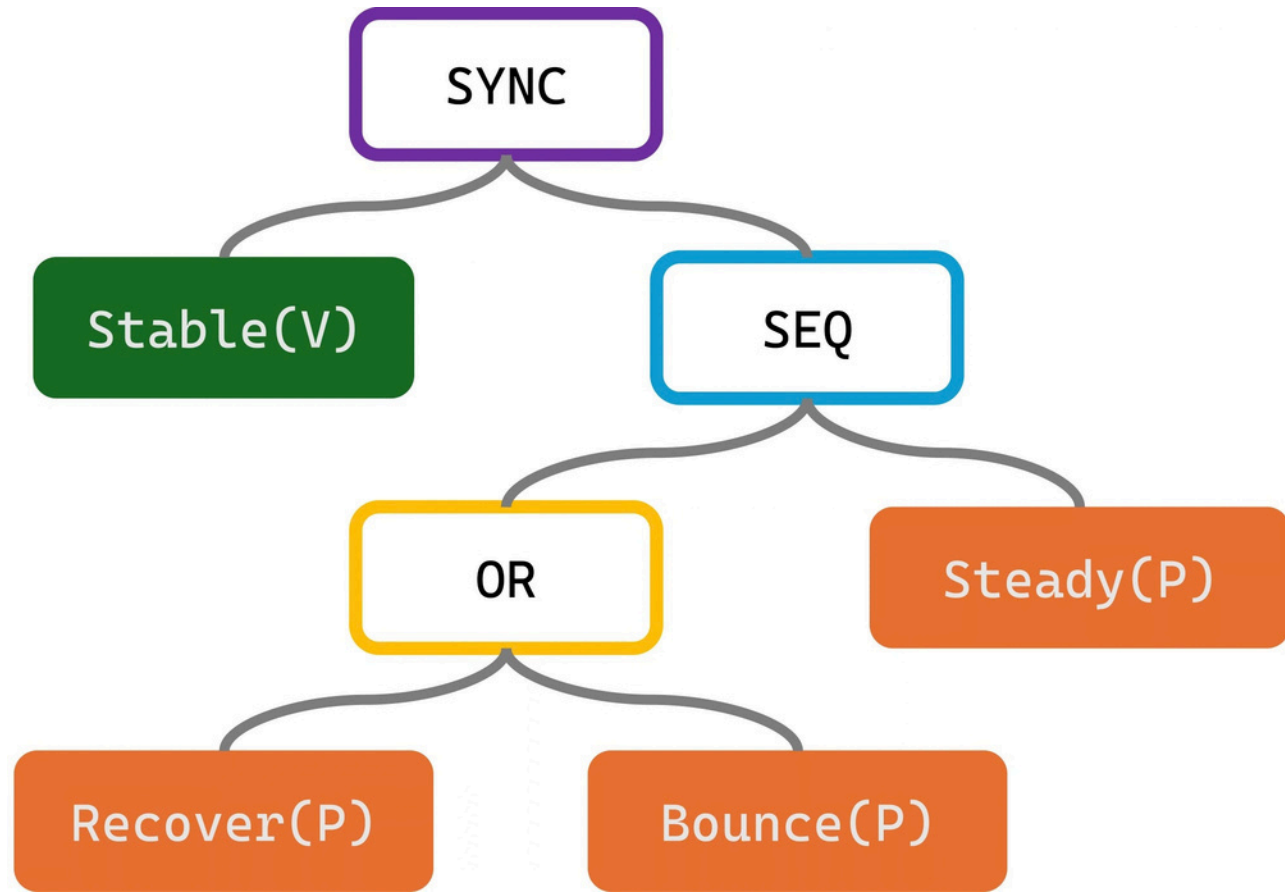
How SELA turns language into event evidence

Text + signals → event class, interval and an inspectable proof tree

1. Parse an Event Logic Tree

“During the **Buildup** event, the *Volume* should remain **stable** across the entire phase. Simultaneously, the *Pressure* should either **recover** slightly with the rate of change decreasing or **bounce** up quickly, and then reach a **steady** phase.”

Logic Analyst: compile the description into a reusable ELT schema.



SEQ order · SYNC overlap

GUARD containment · OR alternatives

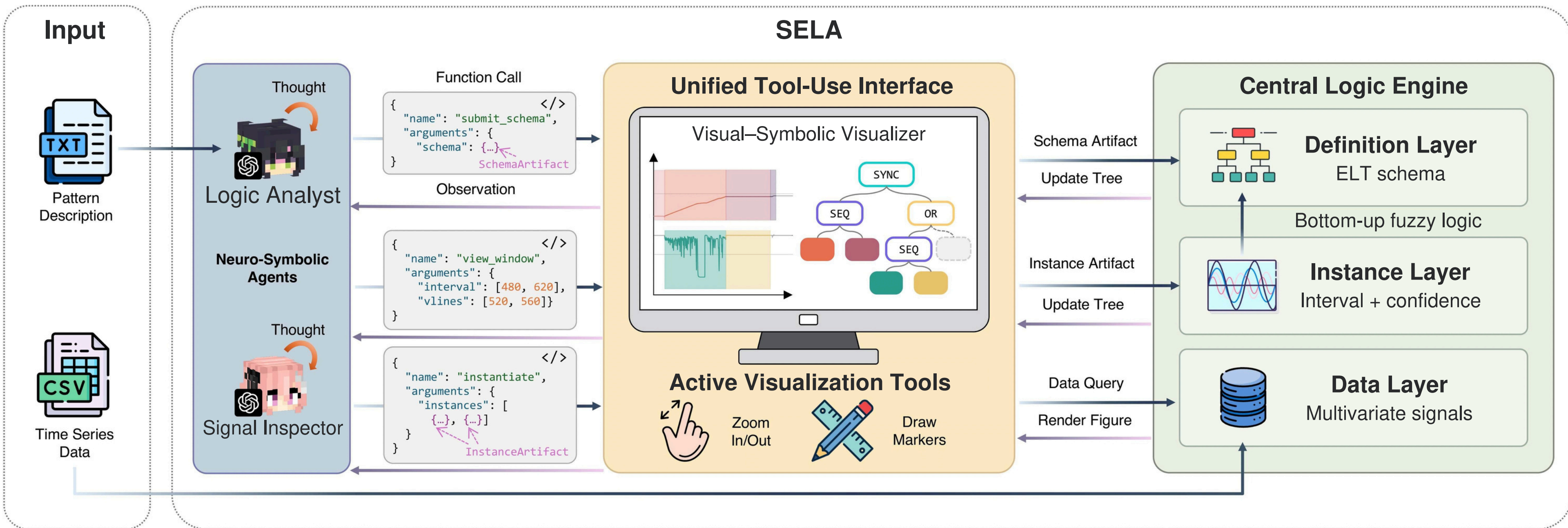
2. Ground the primitives

The **Signal Inspector** uses zoom and markers to ground each primitive with an interval and confidence, then revises it through visual-symbolic feedback.

Separate definition from grounding; return visual feedback after each update.

3. Compose the evidence

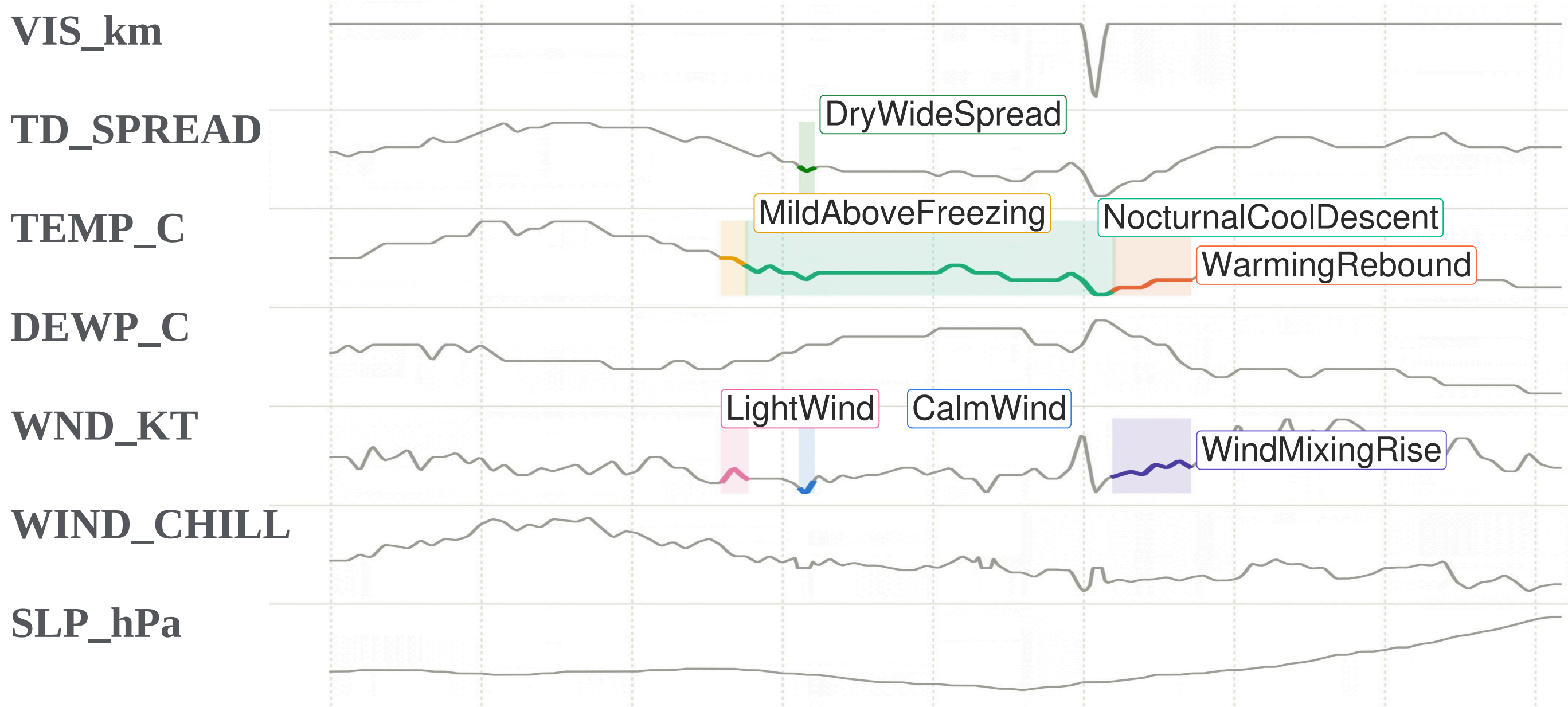
The **Logic Engine** rejects invalid temporal compositions and propagates fuzzy confidence bottom-up. Normalized root scores compare event hypotheses.



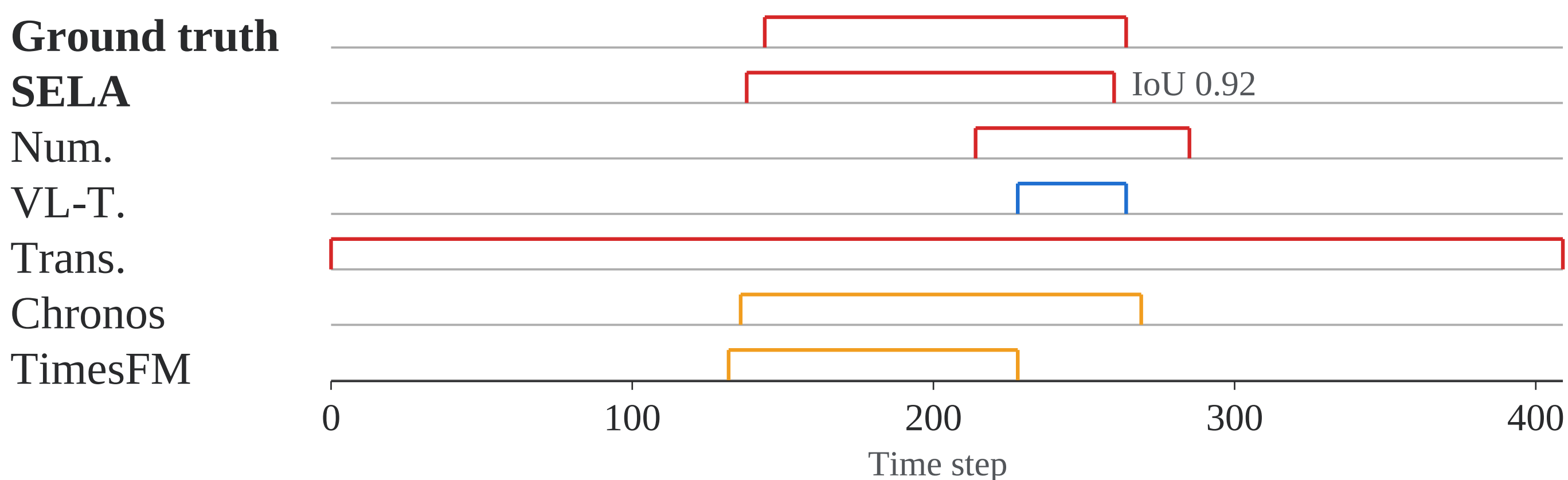
Case study: Frost Freeze—from grounded signals to an event proof

Signals & primitive instances

7 channels · normalized [0, 1]

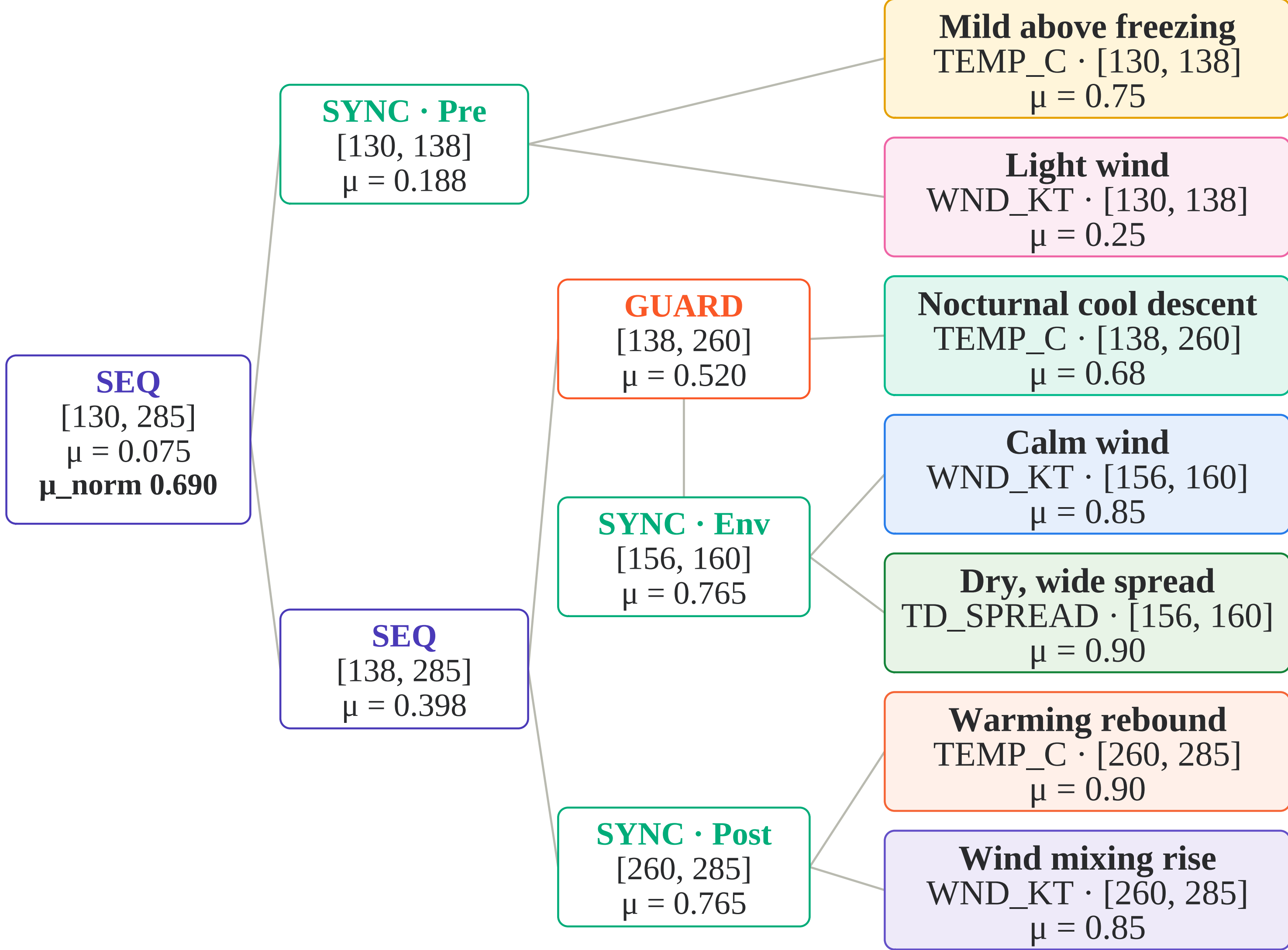


Selected detections · 0/1 · LLM/VLM: GPT-5



Instantiated ELT · intervals and confidence

Grounding example · GPT-5



Red: frost freeze · Blue: dense fog · Orange: extreme cold/wind chill.